

Pools, Outbuildings and the Spacing Numbers

The outdoor half of the topic, where the numbers are the answer and the logic behind each one is easy to reconstruct.

Around water, every rule is either about height (keep conductors out of reach) or about bonding (make everything conductive share one voltage so there is nothing to shock between).

The numbers here

Service drop wires above the pool water	22 1/2 ft
Service drop horizontally from the pool water, in all directions	10 ft
All metal within this distance horizontally must be bonded to the pool grid	5 ft
Pool equipotential bonding conductor	8 AWG solid copper

22.5 feet vertical is set high enough that a pool skimmer pole cannot reach a conductor. Five feet horizontal and twelve vertical is the envelope in which everything conductive joins one grid, so there is no voltage difference for a swimmer to bridge.

YOU HAVE MISSED THIS

Electric service drop wires should be at least ____ feet above the pool water, and at least 10 feet horizontally in all directions.

22 1/2.

YOU HAVE MISSED THIS

An ____ AWG bonding conductor should be attached at the terminal on the exterior of the pool pump.
Eight.

YOU HAVE MISSED THIS

The primary purpose of a GFCI circuit at the pool equipment is:
To protect the pool light.

The pool light sits underwater on a circuit that people are in the water with. That is the specific thing the GFCI at the equipment is there for.

Detached buildings

YOU HAVE MISSED THIS

All of the following conditions violate electrical safety standards in detached garages EXCEPT:
Exterior service wire.

An exterior service wire feeding a detached garage is normal construction, not a violation. Recall from lesson 3 that such a building, fed by a feeder, needs its own grounding electrode.

The service drop clearance ladder

The overhead wires coming to the house have one clearance for every kind of thing underneath them, and the ladder is ordered by what could reach up and touch them. Learn it as one sequence rather than five separate facts.

The numbers here

Above grade or a walkable area	10 ft
Above a driveway	12 ft
Above a public road	18 ft
Clear of windows, doors, porches and decks	3 ft

Ten feet clears a person, twelve clears a car, eighteen clears a truck. The three-foot rule is different in kind: it is horizontal reach, keeping the conductors out of arm's length of anywhere a person can stand.

The spacing numbers indoors

The numbers here

Receptacle spacing — no point along a wall further than this from one	6 ft (so receptacles every 12 ft)
Three-way switch required for a stair with this many risers or more	6
Electric range	40 amp
Electric water heater	240 V / 30 amp
Gas closet dryer / refrigerator	20 amp

Three questions in this topic answer "six"

Receptacle spacing (six feet to the nearest one), the three-way switch stair rule (six risers), and the disconnect rule (six hand movements). Different units, same word. Know which "six" a question is asking for before you answer it.

Say it back

Cover the page. Answer each one out loud.

1. Give the two pool clearance numbers and say what each one protects against.
2. How far horizontally from pool water must metal be bonded, and with what size conductor?
3. What is the GFCI at the pool equipment specifically protecting?
4. What is normal, rather than a violation, in a detached garage?
5. State the receptacle spacing rule in both of its forms.
6. Name the three questions in this topic whose answer is six.
7. Amps for a range, a water heater, and a gas closet dryer.